Credit Card Fraud Detection Using Machine Learning

Nicholas Toma

December 2025

Introduction

This is a personal project exploring the prediction of credit card fraud using machine learning. Credit card fraud refers to the unauthorized use of someone's credit card or card information to make purchases, withdraw cash, or open new accounts. The goal of this project is to build comfort applying machine learning methods to real data in Python, and to practice analyzing and interpreting the resulting models. The full code for this analysis is included at the end of the project file.

Data

The model is trained on credit card transaction data sourced from Kaggle. The dataset contains transactions made by European cardholders, and includes 7,195 fraudulent transactions out of 100,000 total (roughly 7%). The Fraud variable is a binary label equal to 1 if the transaction was fraudulent and 0 if it was legitimate. The dataset also includes cardholder information such as age, gender, bank, and country of residence, as well as transaction-level details such as a unique transaction ID, type of transaction, date, amount, time elapsed since the previous transaction, shipping address, and country where the transaction occurred. From the residence, shipping, and transaction country fields, we constructed binary mismatch variables, since a mismatch between these locations can be a strong indicator of fraudulent activity.

Models

Before modeling, the data was split into stratified 80/20 train and test sets using scikit-learn's train_test_split. Stratification was used because the dataset is imbalanced, so it ensures fraud cases appear in the same proportion in both the training and test sets. Features were then standardized with a StandardScaler fit only on the training data, and two classifiers, logistic regression and KNN, were tuned using stratified 5-fold grid search. The grid search scored candidates using average precision rather than accuracy, since this metric better reflects performance on the rare fraud class by focusing on the precision–recall trade-off.

Logistic Regression

We used scikit-learn's logistic regression model. The grid search tuned regularization strength, which controls how strongly coefficients are penalized and lets the search balance underfitting vs overfitting, along with the penalty type (l1 vs l2). L1 encourages sparse coefficients, effectively performing implicit feature selection, while l2 shrinks coefficients more smoothly. Testing both lets the data determine which structure fits better. The solver was fixed to saga, since it supports both l1 and l2 penalties and scales well to larger datasets, and it was paired with a high max_iter to ensure reliable convergence. Several class weight options were also tested, including balanced, which assigns a much higher penalty for missed fraud cases than for misclassified legitimate ones, directly counteracting the class imbalance.

K-Nearest Neighbors (KNN)

We used scikit-learn's KNN model. The grid search tuned the number of neighbors, since smaller values make the model more flexible and sensitive (lower bias, higher variance) while larger values smooth predictions and reduce noise (higher bias, lower variance). Testing this range allowed the search to find a good balance. The weighting scheme was also tuned, where distance weighting lets closer neighbors have more influence on a prediction, which can help when nearby points are more reliable indicators of class.

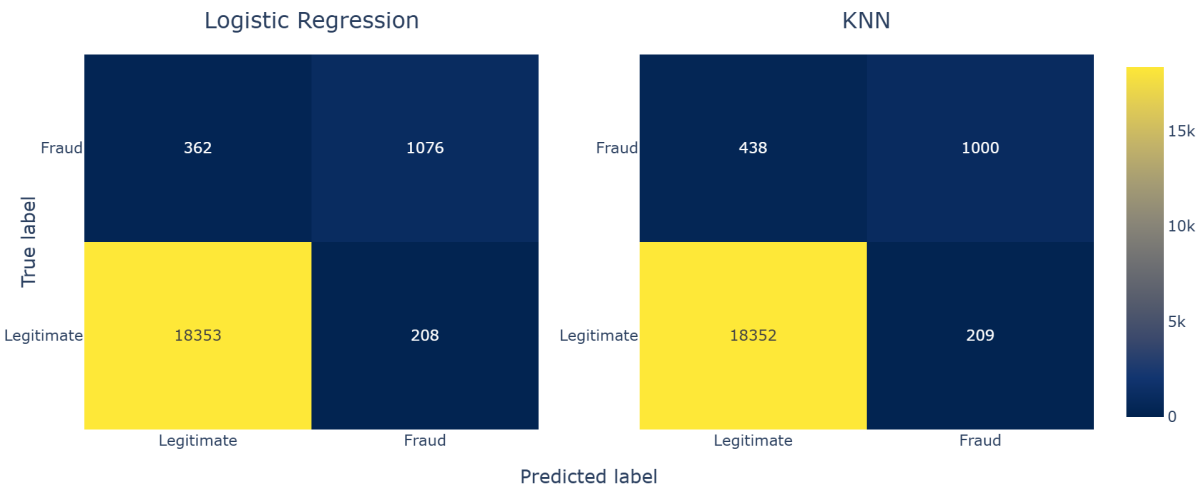
Model Evaluation

With both models tuned, we now assess how well they actually perform at detecting fraud. We evaluate each model using confusion matrices, ROC and precision-recall curves, and feature importance plots, since accuracy alone would be misleading given how imbalanced the data is.

Confusion Matrix

Figure 1 shows the confusion matrices for both models on the test set. Logistic regression correctly identified 1,076 of the 1,438 fraudulent transactions (about 75% recall), while producing 208 false alarms out of roughly 18,561 legitimate transactions. KNN performed slightly worse on the fraud class, correctly flagging 1,000 fraud cases (about 70% recall) with a similar number of false alarms (209). Both models therefore keep the false positive rate very low, but logistic regression catches a somewhat larger share of the actual fraud.

Figure 1: Confusion Matrices



Precision-Recall & AUC ROC

Figure 2 shows the precision-recall and ROC curves for both models. Logistic regression achieved a slightly higher average precision (0.893) than KNN (0.861), meaning it maintains stronger precision as recall increases, which is the more meaningful comparison here given how rare fraud is in the data. Both models also achieved very high ROC-AUC scores (0.989 for logistic regression and 0.984 for KNN), indicating that both are highly effective at ranking fraudulent transactions above legitimate ones overall. Since ROC-AUC can look overly optimistic on imbalanced data, average precision is the more trustworthy metric for comparing the two, and by that measure logistic regression is the stronger model.

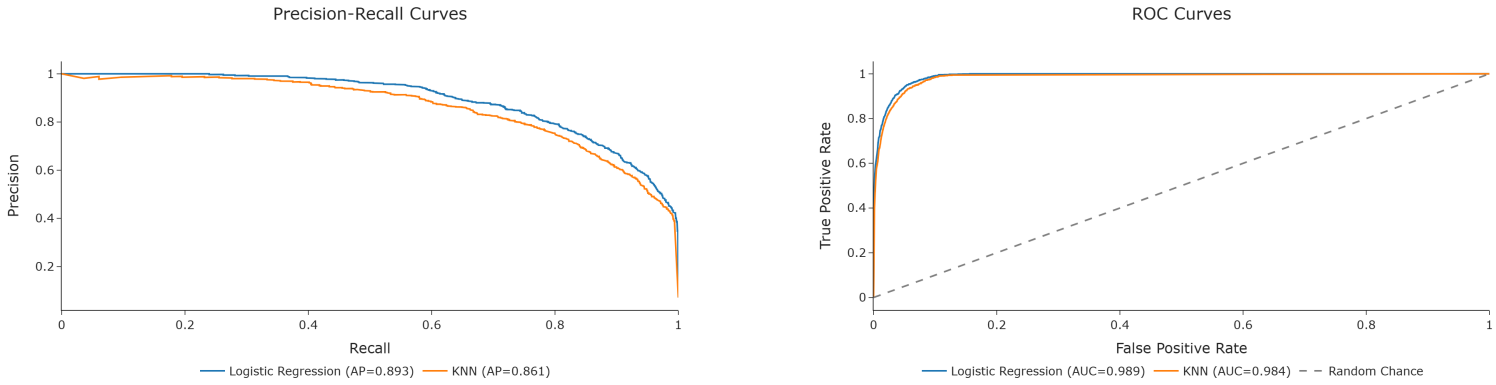


Figure 2: PR & ROC Curves

Feature Importance

Feature importance the logistic regression model was taken from the absolute values of the fitted coefficients, and since KNN has no coefficients, feature importance was instead estimated using permutation importance on a resampled subset of the data. Figures 3 shows the top features driving each model's predictions. For KNN, the transaction country mismatch, time, and shipping/residence mismatch variables were the most influential, followed by age, transaction amount, and country of transaction. For logistic regression, the shipping/residence mismatch and transaction country mismatch variables were by far the strongest predictors, well ahead of transaction amount, PIN entry mode, and time. In both models, the location mismatch variables we engineered rank among the most important predictors, which supports our original hypothesis that a discrepancy between a cardholder's residence, shipping address, and transaction location is a strong signal of potential fraud.

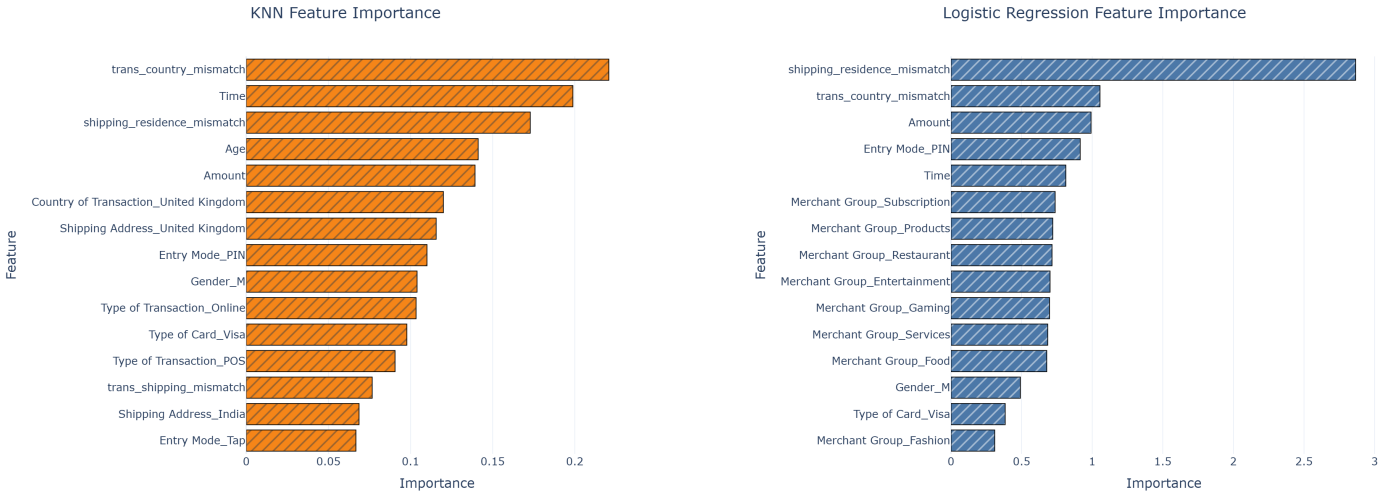


Figure 3: Feature Importance

Conclusion

Overall, both models performed well at distinguishing fraudulent transactions from legitimate ones, achieving high ROC-AUC scores and reasonably strong average precision despite the significant class imbalance in the data. Logistic regression slightly outperformed KNN on the metrics that matter most for this problem - most notably, average precision and fraud recall, while both kept a comparably low false positive rate. This suggests that the relationship between the engineered mismatch features and fraud is close to linear, which logistic regression is well suited to capture.

The feature importance results were one of the most encouraging parts of this project. The location mismatch variables we constructed from the residence, shipping, and transaction country fields turned out to be among the most predictive features in both models. Other variables like Amount, time, and entry mode also contributed meaningfully, which suggests

that fraud in this dataset tends to follow identifiable behavioral patterns rather than just being random.

In terms of practical takeaways, logistic regression's combination of strong performance and interpretability makes it the more attractive model for this use case, especially in a setting like fraud detection where understanding why a transaction was flagged matters. Overall, this project was a useful exercise in applying the full machine learning workflow, from data preparation and model tuning to evaluation, on a realistic and challenging classification problem.